**Ex.No.3**

**Program :**

```
#include <stdio.h>
int main()
{
    int A[100][4];
    int i, j, n, total = 0, index, temp;
    float avg_wt, avg_tat;

    printf("Enter number of process: ");
    scanf("%d", &n);

    printf("Enter Burst Time:\n");
    for (i = 0; i < n; i++)
    {
        printf("P%d: ", i + 1);
        scanf("%d", &A[i][1]);
        A[i][0] = i + 1;
    }

    for (i = 0; i < n; i++)
    {
        index = i;

        for (j = i + 1; j < n; j++)
            if (A[j][1] < A[index][1])
                index = j;

        temp = A[i][1];
        A[i][1] = A[index][1];
```

```
    A[index][1] = temp;

    temp = A[i][0];
    A[i][0] = A[index][0];
    A[index][0] = temp;
}

A[0][2] = 0;

for (i = 1; i < n; i++)
{
    A[i][2] = 0;
    for (j = 0; j < i; j++)
        A[i][2] += A[j][1];

    total += A[i][2];
}

avg_wt = (float)total / n;
total = 0;

printf("P\tBT\tWT\tTAT\n");

for (i = 0; i < n; i++)
{
    A[i][3] = A[i][1] + A[i][2];
    total += A[i][3];

    printf("P%d\t%d\t%d\t%d\n", A[i][0], A[i][1], A[i][2], A[i][3]);
}
```

```
    avg_tat = (float)total / n;

    printf("Average Waiting Time= %f", avg_wt);
    printf("\nAverage Turnaround Time= %f", avg_tat);

    return 0;
}
```

**Output**

```
Enter number of process: 4
Enter Burst Time:
P1: 4
P2: 6
P3: 7
P4: 5
P BT WT TAT
P1 4 0 4
P4 5 4 9
P2 6 9 15
P3 7 15 22
Average Waiting Time= 7.000000
AverageTurnaround Time= 12.500000

=== Code Execution Successful ===
```